

YOLO Model Testing Report

Model:	yolo12m
Model Size:	39.0 MB
Parameters:	20.2 M
FLOPs (640x640):	68.08 GFLOPs
Run Name:	yolo12m_testing_20251124_101029
W&B Run Name:	yolo12m_bdd100k_yolo_test_20251124_101029
Timestamp:	2025-11-24 10:18:42
Dataset:	bdd100k_yolo - test split
Images Processed:	20000
Total Execution Time	469.67s
IoU Threshold:	0.5

Inference Performance

Metric	Value
Average Inference Time	20.24 ms per image
FPS (Frames Per Second)	49.41

Note: Inference time includes preprocessing, model inference, and postprocessing.

Overall Accuracy Metrics

Metric	Value
Precision	0.2867
Recall	0.6503
F1-Score	0.3979
mAP@0.5	0.1009

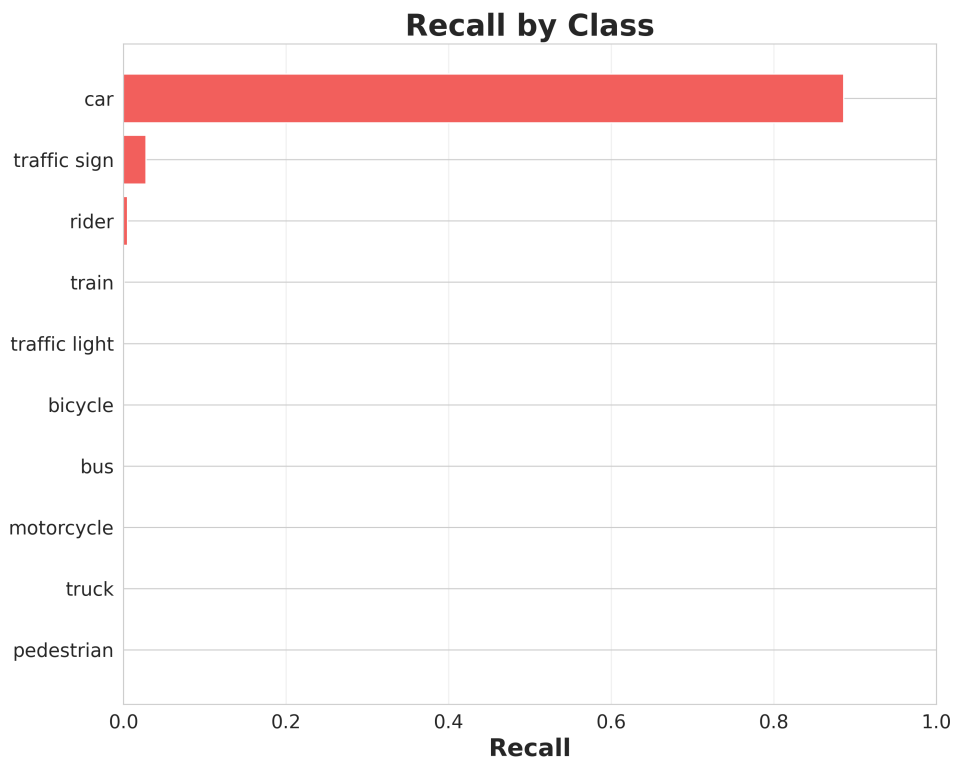
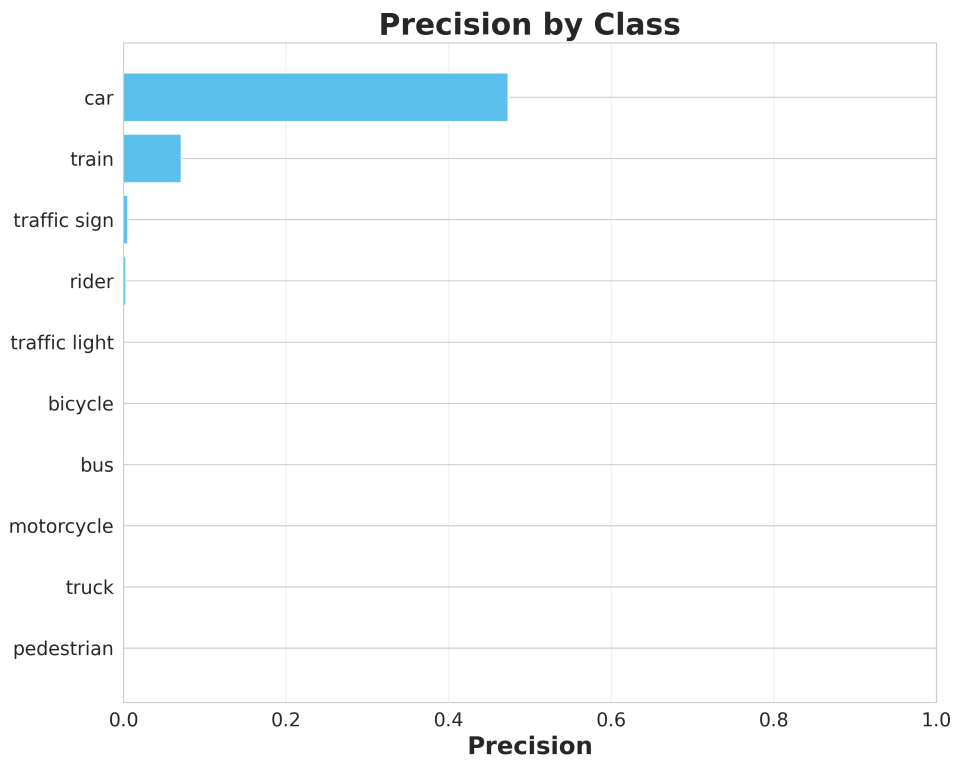
mAP@0.5:0.95	0.0598
True Positives	97508
False Positives	242633
False Negatives	52441

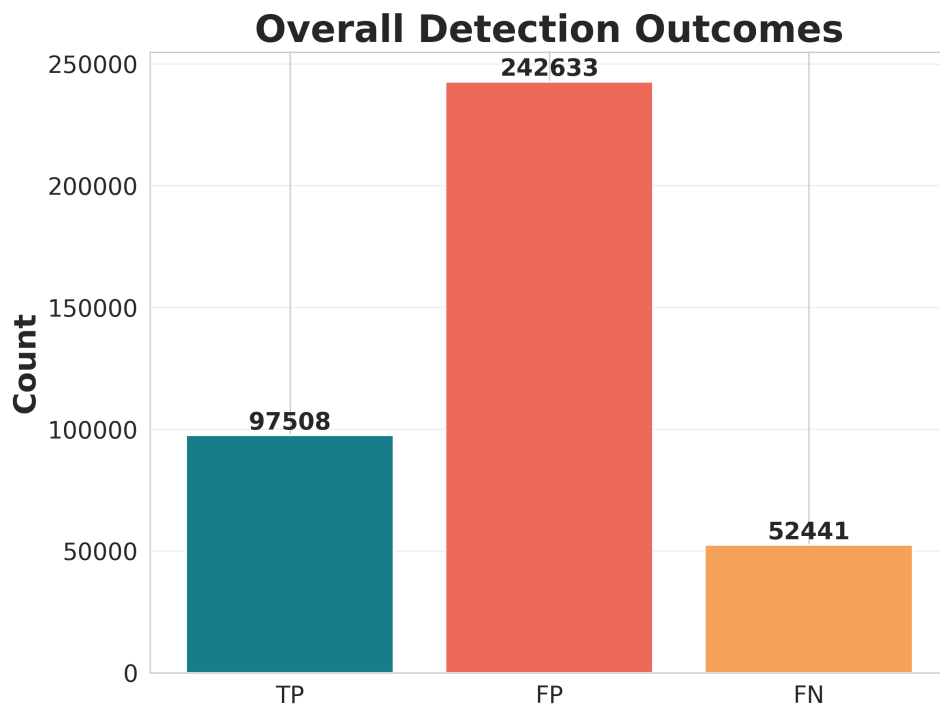
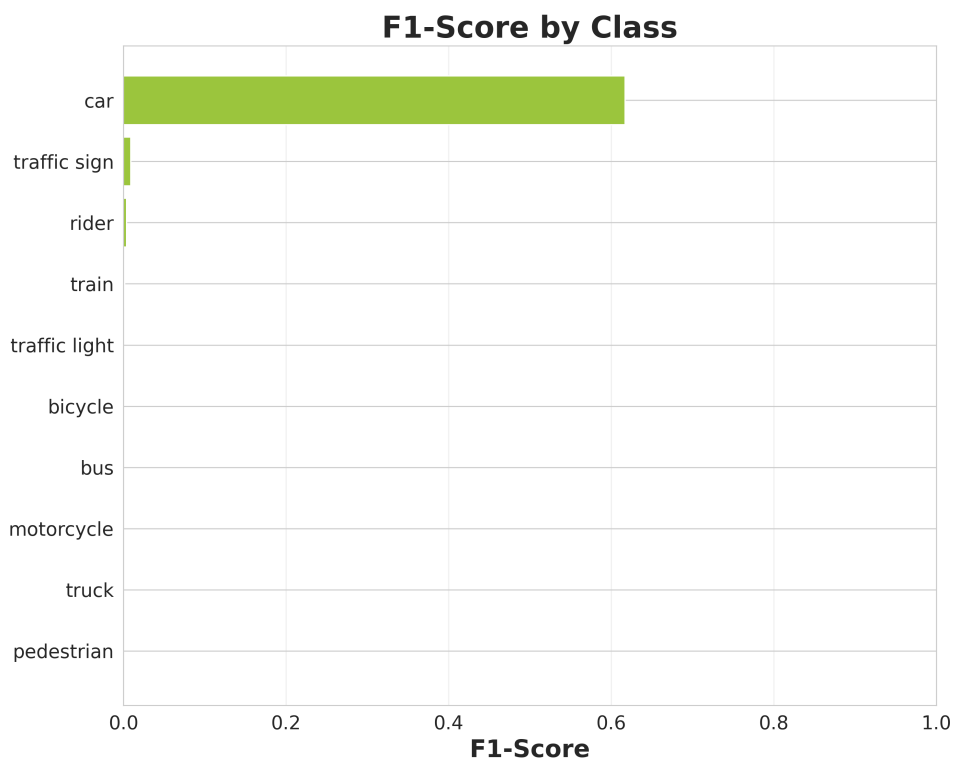
Confusion Matrix

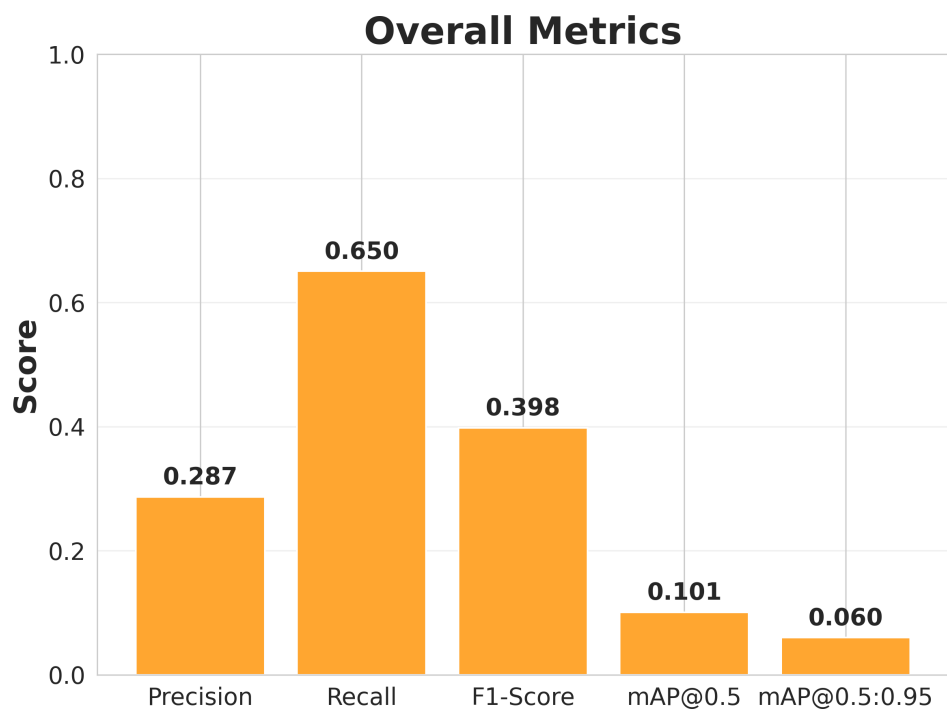
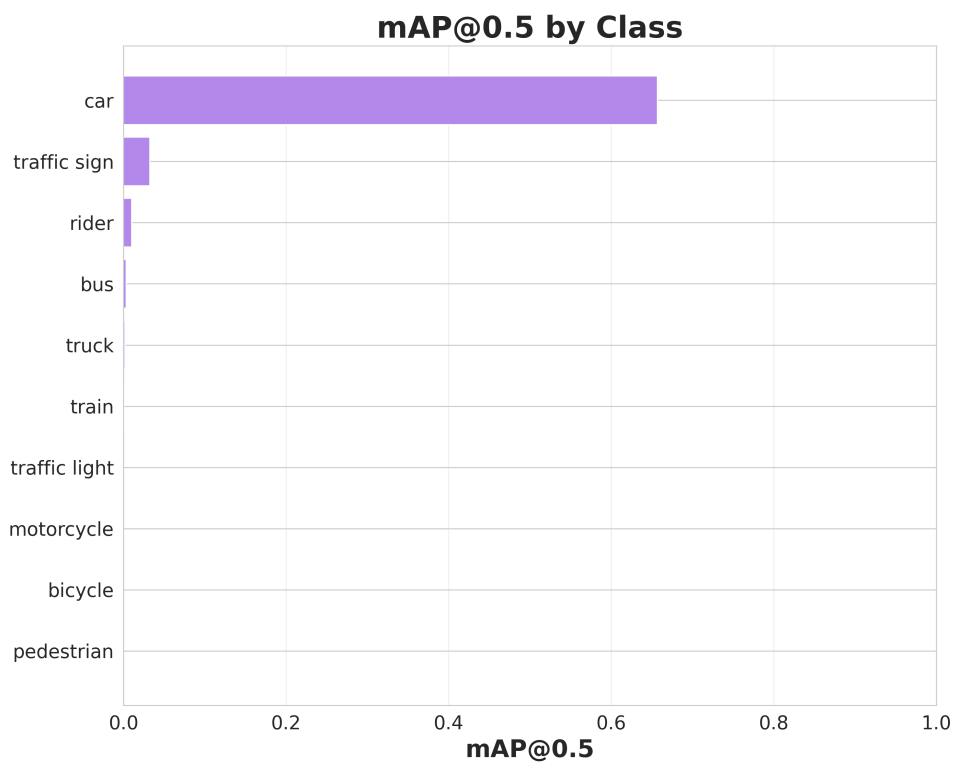
Confusion Matrix (yolo12m validation)											
True Class	pedestrian	628	73	4					4	16	
	rider	4									
	car	6	97117	908	179				19	52	
	truck	16	13							1	
	bus										
	train		450	403	1330	2				2	
	motorcycle		6	5	13	6					
	bicycle	1	3295	3709	371	2			2	11	
	traffic light										
	traffic sign		6	1		2			9474	385	
		pedestrian	rider	car	truck	bus	train	motorcycle	bicycle	traffic light	traffic sign
Predicted Class											

Correct Predictions (Diagonal Sum): 97508
Total Matched Predictions: 373138

Performance Visualizations







Per-Class Performance

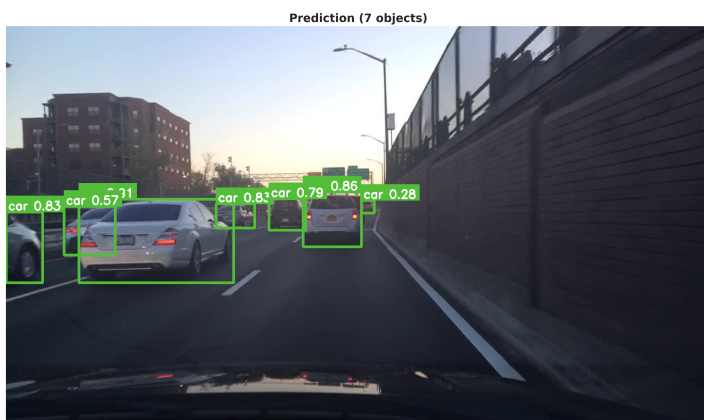
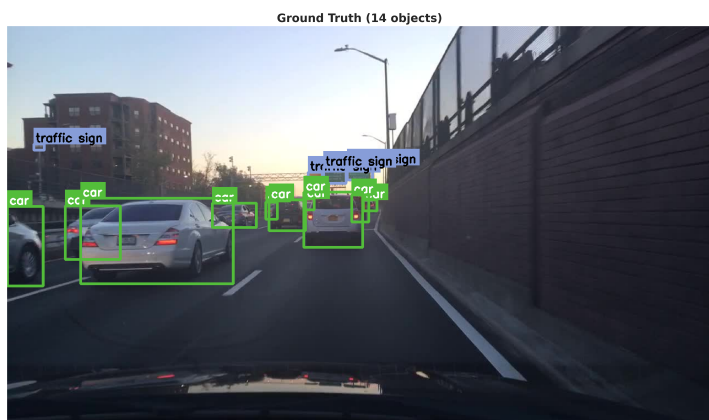
Class	TP	FP	FN	Precision	Recall	F1-Score	mAP@0.5
pedestrian	0	0	12354	0.0000	0.0000	0.0000	0.0000
rider	4	1289	754	0.0031	0.0053	0.0039	0.0104
car	97117	107976	12463	0.4735	0.8863	0.6173	0.6571
truck	0	8701	447	0.0000	0.0000	0.0000	0.0019
bus	0	3217	6	0.0000	0.0000	0.0000	0.0037
train	2	26	2782	0.0714	0.0007	0.0014	0.0003
motorcycle	0	0	115	0.0000	0.0000	0.0000	0.0000
bicycle	0	0	10088	0.0000	0.0000	0.0000	0.0000
traffic light	0	52817	9	0.0000	0.0000	0.0000	0.0000
traffic sign	385	68607	13423	0.0056	0.0279	0.0093	0.0327

Sample Predictions: Ground Truth vs Model

Sample #1: d5e37340-1ac1ba8d.jpg

Ground Truth Objects: 14 | Predicted Objects: 7

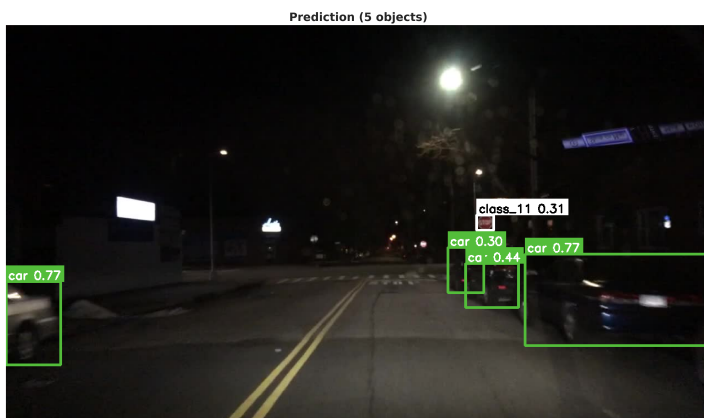
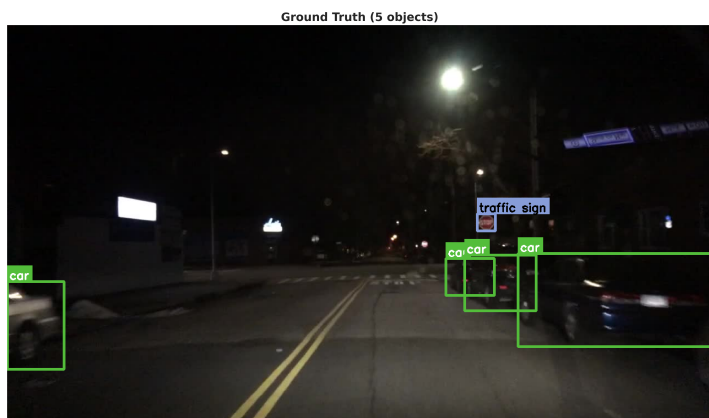
Weather: clear | Scene: highway | Time of Day: dawn/dusk



Sample #2: dcb29cf6-6b43a818.jpg

Ground Truth Objects: 5 | Predicted Objects: 5

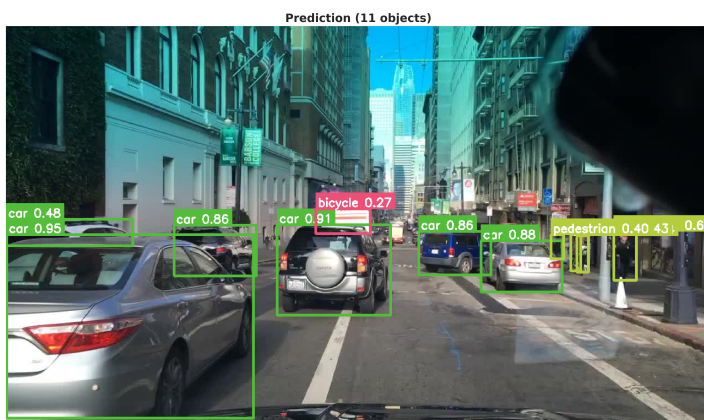
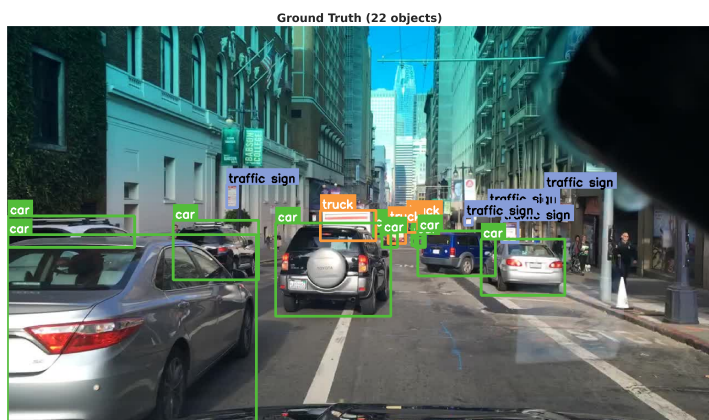
Weather: snowy | Scene: city street | Time of Day: night



Sample #3: d772c580-9eaf06bf.jpg

Ground Truth Objects: 22 | Predicted Objects: 11

Weather: clear | Scene: city street | Time of Day: daytime



Sample #4: d4d8e3b4-743aa8b2.jpg

Ground Truth Objects: 28 | Predicted Objects: 27

Weather: overcast | Scene: city street | Time of Day: daytime



